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FOUNDATIONAL TIME SERIES MODELS FOR EMERGENCY DEPARTMENT OCCUPANCY FORECASTING

Bachelor's Thesis
Faculty of Information Technology and Communication Sciences
Examiner: M.Sc. Eetu Pulkkinen
December 2025

ABSTRACT

Valtti Järvi: Foundational time series models for emergency department occupancy forecasting
Bachelor's Thesis
Tampere University
Bachelor's Programme in Computing and Electrical Engineering
December 2025

Emergency department overcrowding is a significant threat to patient safety..

Keywords: time series forecasting, foundational models, emergency department occupancy prediction

The originality of this thesis has been checked using the Turnitin OriginalityCheck service.

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Valtti Järvi:
Kandidaatin tutkielma
Tampereen yliopisto
Tieto- ja sähkötekniikan kandidaattiohjelma
Joulukuu 2025

Avainsanat: keyword, keyword, keyword, keyword, keyword

Tämän julkaisun alkuperäisyys on tarkastettu Turnitin OriginalityCheck -ohjelmalla.

CONTENTS

1. Introduction	1
2. Background	3
2.1 Emergency department	3
2.1.1 Definition of crowding	3
2.1.2 Consequences of crowding	3
2.1.3 Emergency department forecasting	4
2.2 Time series forecasting	4
2.3 Foundational models	5
2.3.1 Pre-training and fine-tuning paradigm	6
2.3.2 Selected foundational time series models	6
3. Methodology	7
3.1 Data	7
3.2 Experimental setup	7
3.3 Evaluation	7
3.4 Models	8
3.4.1 Chronos	8
3.4.2 TimesFM	9
3.4.3 Moirai	9
3.4.4 Sundial	10
3.4.5 TiRex	10
3.4.6 Benchmark models	10
4. Results	12
4.1 Predictive performance	12
4.2 Effect of context length	13
4.3 Effect of covariates	13
4.4 Computational cost	13
5. Discussion	15
5.1 Limitations	15
6. Conclusion	16
7. References	17
Appendix A: Computational costs for multivariate forecasts	21

1. INTRODUCTION

Emergency department (ED) crowding has been widely documented to negatively impact a range of clinically relevant patient outcomes [1], including higher mortality rates [2, 3] and longer lengths of stay [4]. Accurate predictions of future ED occupancy levels would enable proactive resource allocation aiming to alleviate or even prevent future crowding events.

Emergency department forecasting refers to the process of predicting the future status of the ED, typically using timeseries forecasting models. Traditionally, these have been trained using historical utilization data – sometimes with external variables such as weather and traffic data [5, 6]. However, training machine learning models involves a number of problems. First, every hospital has unique characteristics. A model trained with data from one hospital may not accurately predict occupancy levels for other hospitals. This means that a new model must be trained for each hospital at the time of implementation and repeatedly throughout the production cycle, which can be a computationally heavy and time-consuming task. Second, accessing the required data for each hospital can be a challenge due to either regulatory or practical reasons.

Foundational models (FM) are (tähän kuvaus siitä mitä ovat). In the context of emergency department forecasting, they offer a set of unique advantages in comparison to traditional machine learning algorithms. First, (ei uudelleen koulutustarvetta), Second (voi ottaa suoraan käyttöön), Third (onko joku kolmas vielä?

Foundational models do not need to be trained on task-specific data. If they can predict ED occupancy accurately enough, the difficult task of model training can be skipped. This could save a substantial amount of computation and man-hours.

This thesis evaluates whether the newly introduced foundational time series models can be used to predict ED utilization of Tampere University hospital ED "Acuta". A public dataset (viite) of the total hourly occupancy of the ED from x to y is used as the target variable and the performance of the foundational models are evaluated in comparison to the performance of state of the art models documented by [5].

The research questions of this thesis are as follows: (1) how does the performance of the foundational models compare to the traditional SOTA models; 2) are there differences between the performances of the existing FM implementations, and 3) how do the FM

models differ in terms of computational requirements?

The thesis is structured as follows. Chapter 2 provides the necessary background on emergency departments, crowding, time series forecasting, and foundational models. Chapter 3 details the experimental setup and the dataset used. Chapter 4 presents the empirical results, and Chapter 5 discusses the implications of these results in the context of existing knowledge. Finally, Chapter 6 provides final conclusions that can be drawn from the work.

2. BACKGROUND

2.1 Emergency department

Tähän muutamalla lauseella päivystyksen määritelmä

2.1.1 Definition of crowding

Emergency departments (EDs) serve a critical role in healthcare systems by providing immediate access to medical services [1]. Worldwide, they function as a crucial 'safety net,' addressing acute health problems and responding to local and national disasters [7, 8]. In the United States, more than half of all inpatient hospital admissions originate from the ED [7].

ED crowding does not have a single widely accepted definition. The proposed frameworks include NEDOCS or EDWIN but their utilization and practicality is uncertain. Regardless of the definition, crowding is always

a situation in which demand for service exceeds the ability to provide care within a reasonable time, causing physicians and nurses to be unable to provide quality care

as was eloquently expressed by [9].

2.1.2 Consequences of crowding

Regardless of the definition, crowding has become a worldwide public health issue [10], the negative impacts of which can be divided into three broad categories based on their target: the patient, the staff, and the entire healthcare system. First, crowding has been repeatedly associated with increased mortality and increased number of medication errors [11](tarkista). Second, crowding has been shown to increase the stress levels of nurses and physicians [12]. Third, crowding results in ambulance diversions diverting ambulances to another ED [7], which delays patient access to care and disrupts coordination between hospitals. In addition, crowding compromises the ability of the ED to respond to mass emergencies or at least their capability to provide quality care to less urgent patients [10].

2.1.3 Emergency department forecasting

Emergency department forecasting refers to the process of predicting future utilization of the ED. Accurate forecasting would enable proactive resource allocation that could ultimately prevent overcrowding along with its harmful consequences. In previous literature, the predictability of a wide set of target variables has been investigated. These include the number of patient arrivals, occupancy, patient admission, length of stay, or occupancy ratio.[13]

Tähän kuuluu osio, jossa viitataan kunkin vastemuuttujan osalta tiettyyn juttuun (katso tuosta Gul et al). Sitten kerrotaan, että mitkä on kirjallisuuden puutteet. Lyhyimmillään: "The utilization of foundational models has previously only been reported by (viite

2.2 Time series forecasting

The goal of time series forecasting is to predict the future evolution of a data sequence based on its past values. External data sequences can be included for more accurate forecasts. [6] For example, future hourly ED occupancy can be predicted using past ED occupancy data, calendar information, and weather.

A discrete-time time series is a set of discrete values $\{x_t\}$, where t denotes a specific time [14]. In this thesis, the term 'time series' refers always to discrete-time time series. The intervals between t are usually fixed and of equal length.

The term univariate refers to data that consists of a single attribute or variable. In contrast, multivariate data consists of multiple related variables observed over time, where the relationships between variables can provide additional information for forecasting. It can be denoted as $\{\mathbf{x}_t\}$, where $\mathbf{x}_t$ is a vector. [14] Other variables than the one we predict are called covariates or exogenous inputs.

For a time series to be suitable for forecasting, it must exhibit identifiable patterns. The trend component represents the long-term direction or gradual change in the data, which, in the case of ED occupancy data, may reflect factors such as population growth. Seasonality means repeating patterns that occur at regular intervals. For example, ED occupancy peaks daily between 16:00-19:00 and is at its lowest between 04:00-07:00 [5]. Random noise refers to irregular and random fluctuations that cannot be explained by any pattern. After all, many things, such as accidents happen randomly. These and many other patterns, including patterns from covariates, make forecasting possible. [14]

A *model* is a mathematical equation that describes the patterns found in the data. The objective is to find a model that effectively captures the underlying structure of the time

series. This relationship can be formally expressed with a general equation:

$$y_{t+1} = f(y_t, y_{t-1}, \dots, \mathbf{x}_{t+1}, \mathbf{x}_t, \dots) + \epsilon_{t+1} \quad (2.1)$$

The function f represents the model itself. Specifies the relationship between the target variable y_{t+1} and the past values of the target variable $y_t, y_{t-1}, \dots$, and the present and past values of the covariates $\mathbf{x}_t, \mathbf{x}_{t-1}, \dots$. The term ϵ_t is the random noise that is not explained by the model. The function f can be a simple statistical model or a more complex machine learning model like a neural network. [6]

The future is always unpredictable to some extent. Therefore, predicting the entire probability distribution of a future value, rather than just a single value, is often useful. This is called a forecast distribution. Prediction intervals represents a range of values that the future value could take given some probability. For example, a 90 % prediction interval contains the actual value with 90 % probability. In contrast, a point forecast represents the best single estimate of a future value. It is often the mean or median of the forecast distribution. [6]

2.3 Foundational models

Foundational models are artificial intelligence (AI) systems that perform well in a wide range of diverse tasks. These models are trained on massive amounts of diverse data. Foundational models use deep neural network architectures with parameters ranging from hundreds of millions to hundreds of billions, depending on the application domain. [15] For example, foundation models have been developed for time series forecasting. A single foundation time series model can predict diverse time series data, from ED occupancy to stock prices.

Foundational models differ from traditional machine learning models that are trained for a specific purpose. For example, a model can be trained to forecast ED occupancy for one specific hospital. Advancements in computing have made it possible to train a larger model for multiple different purposes [15].

Transfer learning is a concept of transferring knowledge from one learned task to other tasks [16]. From vast training data, fundamental structures that are true for all tasks in one domain, are learned [16]. For example, ED occupancy and stock prices could have same kind of daily fluctuation. Transfer learning and the vast size of foundation models is what makes them work [15].

In [15], the authors state three advancements that have enabled the size of foundation models. First, improvement in computational hardware. Second, the Transformer architecture that efficiently utilized this hardware for training [17]. Third, access to larger datasets for training.

2.3.1 Pre-training and fine-tuning paradigm

Deep learning is a part of machine learning that uses deep neural networks to learn from data. These networks learn data representations at several levels. [18] Deep learning powers most modern AI models. All foundation models in this thesis use deep learning, as does the TFT baseline [19].

In [20], the author outlines three key aspects of developing machine learning models. First step is choosing representation, or model. In case of deep learning, this is done by choosing a model architecture. Then the model is optimized or trained to fit the data. Training is done by adjusting the model parameters to minimize a loss function. This step is computationally intensive and often requires massive datacenters and large amount of time [15]. The final step is evaluation, where the model's performance is measured using data not seen during training. Deep learning follows this framework [18].

Foundational models use different framework and they represents a paradigm shift from traditional machine learning framework. For a specific task, already pre-trained foundation model can be selected. Foundation model is often fine-tuned for a specific task. After fine tuning, the model is evaluated. [15]

This approach saves a lot of computation, time and effort, since the foundation model is already pre-trained. It democratizes access to advanced AI models. In case of ED occupancy forecasting, every hospital has its own characteristics. Therefore model trained for one hospital would perform poorly for other hospitals. Training a separate model for each hospital would be expensive. Foundational models can solve this problem.

2.3.2 Selected foundational time series models

This section describes the foundational models selected for this study. The models are compared using their architecture and key forecasting parameters. The *context length* is the number of past time steps the model uses to make a prediction. The *forecasting horizon* is the number of future time steps it predicts. All selected models have promising zero-shot capabilities and can be fine-tuned for specific tasks.

3. METHODOLOGY

ED occupancy was used as a forecasting metric by Tuominen et al. [5]. To directly compare results with Tuominen et al., this thesis uses ED occupancy as its crowding metric. All patients from all ED treatment spaces are summed up to this metric. All parts of input-throughput-output model are affected in occupancy, which makes it useful. [5]

3.1 Data

- where the data came from
- describe the data used (variables, time range,)
- how it was processed
- data split for fine tuning

3.2 Experimental setup

- task is to predict 24h occupancy to the future (zero shot)
- details on different context length inputs, uni/multi
- finetuning setup
- hardware setup and library versions

3.3 Evaluation

In this thesis same metrics are used as in Tuominen et al. [5], to enable direct comparison. They used mean absolute error (MAE) and root mean squared error (RMSE) for point forecasts.

MAE is defined as

$$\text{MAE} = \frac{1}{n} \sum_{t=1}^n |y_t - \hat{y}_t|, \quad (3.1)$$

where y_t is the ground truth value and $\hat{y}_t$ is the prediction made by the model. RMSE is

defined as

$$\text{RMSE} = \sqrt{\frac{1}{n} \sum_{t=1}^n (y_t - \hat{y}_t)^2}, \quad (3.2)$$

using the same notation.

MAE measures forecast accuracy in the same units as the data and treats all errors equally [6]. This makes it easy to interpret. A MAE of 5 means forecasts are off by 5 patients on average. RMSE squares the errors before averaging, which gives more weight to larger errors [6]. This makes RMSE more sensitive to outliers and particularly useful when large errors are more harmful than small ones. In ED occupancy forecasting, this property is valuable because large prediction errors could lead to understaffing or overstaffing. This could cause crowding or wasted resources.

3.4 Models

The models for this study were selected based on three criteria: open-source availability, ease of use, and performance on the GIFT-Eval benchmark [21]. All chosen models have promising zero-shot capabilities, can be fine-tuned, and are available for download from GitHub.

-settings and hyper-parameters that were used for each model (mean or median) -when and where downloaded

-short description of baseline models

3.4.1 Chronos

[22] is a family of foundational time series models developed by Amazon Science. It adopts a language modeling framework. This treats time series forecasting as a next-token prediction task. Based on the T5 transformer architecture [16], Chronos processes continuous time series data by scaling and quantizing it into a fixed vocabulary of discrete bins. The model is trained using cross-entropy loss to predict the next token. This frames forecasting as a classification problem at each time step. To generate forecasts, the model predicts one token at a time. Each predicted token is used as input for predicting the next one. These discrete tokens are then dequantized and inverse-scaled to produce continuous forecast values. Multiple forecast paths are sampled to compute quantile and mean forecasts.

The Chronos family includes four models with parameters ranging from 20 to 710 million. All variants have a maximum context length of 512 and a maximum forecasting horizon of 64. The training data was augmented using the TSMIXup and KernelSynth algorithms. TSMixup creates new time series by combining existing ones. KernelSynth generates

synthetic data using Gaussian processes. As of September 2025, Chronos-bolt-base holds fourth place in GIFT-Eval for models with training data overlap to the test set [23].

3.4.2 TimesFM

, developed by Google Research [24], is a decoder-only transformer model. Its architecture consists of input layers, a transformer stack, and output layers. The model uses a technique called patching. Patching divides the input time series into fixed-length segments. The model processes these segments instead of individual time points. This reduces the number of steps required for long-horizon forecasting. Input patches are 32 time steps long. Output patches are 128 time steps long. The model is trained using Mean Squared Error (MSE) loss on point forecasts.

The latest version, TimesFM 2.5, released in September 2025, has 200 million parameters and an extended context length of 16384. The models were pretrained on approximately 100 billion time points from diverse sources. These included Google Trends, Wikipedia pageview statistics, and public datasets such as M4 and Electricity. This data was augmented with synthetic data from ARMA processes. As of September 2025, TimesFM 2.5 holds third place on the GIFT-Eval leaderboard [23].

3.4.3 Moirai

, developed by Salesforce AI Research [25], is a versatile time series forecasting model. The version used in this thesis is Moirai 2.0. It was released on August 8, 2025 [26]. Moirai uses a novel decoder-only transformer architecture designed for heterogeneous multivariate time series. Its key feature is using varying patch sizes based on data frequency. Higher-frequency data uses larger patches. Each patch is processed by a linear layer and assigned a time ID and a variable ID. These IDs inject positional information into the transformer through rotary embeddings and learnable scalars. The transformer's output patches are projected to produce parameters for a mixture of probability distributions. These parameters include the weights for combining the distributions into a final weighted sum. This sum forms the final probabilistic forecast.

Moirai 2.0 has approximately 12.4 million parameters. This is 96% smaller than the original Moirai Large model. Its architecture does not impose a hard limit on context length. Moirai 2.0 was trained on a combination of the GIFT-Eval dataset, synthetic data from KernelSynth and TSMIXup, and internal Salesforce data. As of September 2025, Moirai 2.0 holds the sixth place on the GIFT-Eval leaderboard [23].

3.4.4 Sundial

is a family of models developed at Tsinghua University [27]. It was released on February 2, 2025. It is the first foundational time series model to output probabilistic predictions directly. This is done by producing multiple forecast samples, where mean median and quantile forecasts are calculated. Sundial uses an encoder-only transformer architecture and works directly on continuous values. The model processes time series data in patches of size 16. It is trained using a novel TimeFlow Loss function. A key feature is its non-autoregressive prediction mechanism. It predicts multiple patches at once, enabling millisecond-fast inference.

Sundial has a context length of 2880 but can handle shorter, arbitrary context lengths. The model was trained on TimeBench. This is a large-scale dataset created by the authors, which contains over a trillion data points from real-world sources like weather and finance. As of November 2025, Sundial holds the 10th place on the GIFT-Eval leaderboard [23]. This demonstrates strong zero-shot forecasting capabilities.

3.4.5 TiRex

TiRex was developed by NXAI GmbH in collaboration with academic researchers. It was released on May 29, 2025 [28]. It is the only model in this thesis based on the xLSTM architecture rather than a transformer [29]. TiRex normalizes inputs using z-score normalization and processes data in patches of size 32. Input patches are projected into an embedding dimension by a residual block. This embedding is fed into the xLSTM. A second residual block projects the xLSTM output back into the patch dimension to produce forecasts. The outputs are scaled back to the original target space. The model predicts nine quantiles for each time step.

TiRex has 35 million parameters, making it computationally efficient. Its xLSTM architecture propagates hidden states from patch to patch. This removes hard constraints on context length. A length of 2048 was used in the original paper. The model's training data comprises 47.5 million time series from Chronos's training data, synthetic data similar to KernelSynth, and parts of the GiftEval pre-training dataset. As of November 2025, TiRex holds second place on the GIFT-Eval leaderboard among models with training data overlap to the test set [23].

3.4.6 Benchmark models

Näitä ei tarvitse avata edes tällä tarkkuudella, koska sitten pitäisi avata kaikki, TFT, DeepAR, N-Beats ja ne. Voi vaan listata esim. jokainen omassa paragraphissa tai peräti listassa. Kts. Tuominen2023 referenssiksi.

Seasonal naïve forecasts future values by repeating the last observed value from the last seasonal period. For a time series y_t with seasonal period m , the forecast for time $t + h$ is:

$$\hat{y}_{t+h|t} = y_{t+h-m}, \quad (3.3)$$

where $\hat{y}_{t+h|t}$ is the forecast at time $t + h$, and y_{t+h-m} is the value from the prior seasonal cycle. For seasonality dominated data this is great baseline to compare other models against. [6]

Autoregressive integrated moving average with exogenous inputs (ARIMAX) is a statistical model that combines linear regression and the ARIMA model. The forecast for time t is given by:

$$y_t = \beta_0 + \beta_1 x_{1,t} + \cdots + \beta_k x_{k,t} + \eta_t, \quad (3.4)$$

where $x_{i,t}$ are the covariates, β_i are their coefficients, and η_t is the ARIMA error term. The term η_t uses correlation between y_t and its previous values y_i . This dependence of previous values is called autocorrelation. [6]

LightGBM is a highly efficient implementation of gradient boosting decision tree algorithm [30]. Efficient implementation makes LightGBM effective for large-scale datasets. Temporal fusion transformer (TFT) is deep neural network architecture for time series forecasting with integrated interpretability [19]. Tuominen et al. [5] used these two models to forecast ED occupancy. LightGBM achieved the best results.

4. RESULTS

Otsikot tutkimuskysymysten mukaan

4.1 Predictive performance

Table 4.1

Model	FS	MAE	RMSE
Moirai	A	11.10	13.69
SN	U	9.53	12.58
ARIMA	U	7.78	10.50
Chronos	U	7.25	9.81
TFT	U	7.18	9.58
TFT	A	6.86	9.04
LightGBM	U	6.66	8.95
LightGBM	A	6.63	8.77
Sundial	U	6.57	8.85
TiRex	U	6.51	8.78
Moirai	U	6.35	8.59
TimesFM	U	6.29	8.45

Table 4.1. Best MAE and RMSE for each model. Foundational models are denoted with bold font. FS denotes Feature Set, where U is univariable and A is all covariates.

4.2 Effect of context length

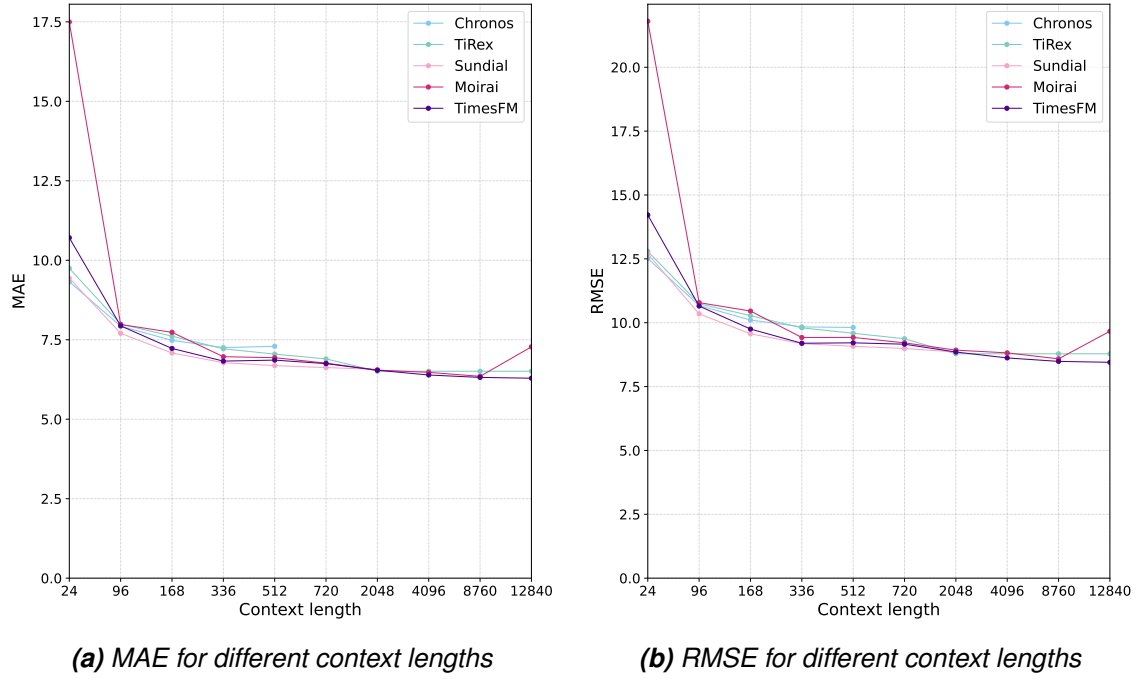


Figure 4.1. MAE and RMSE for different context lengths. All models take univariate input.

4.3 Effect of covariates

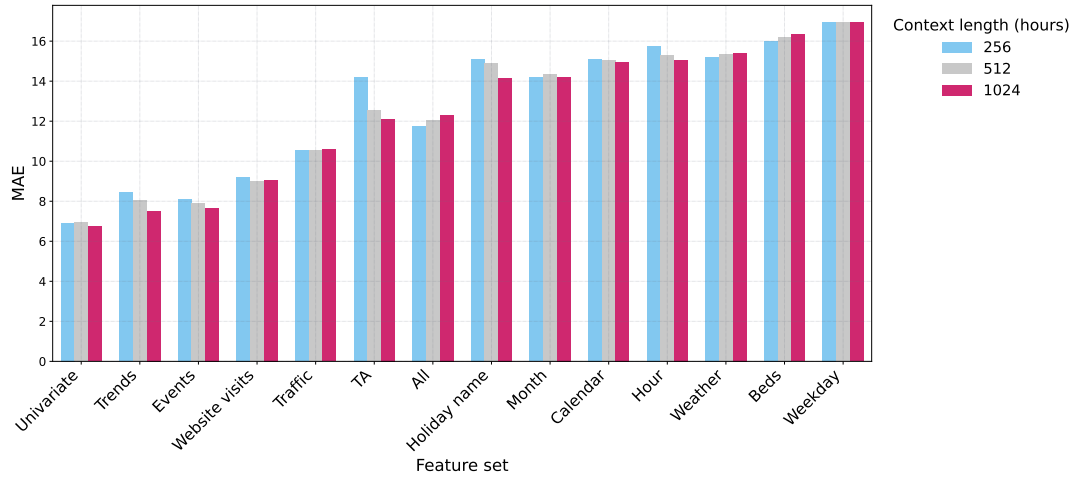


Figure 4.2. MAE across different multivariate feature sets. The "All" category includes all covariates, while other groups represent specific covariate subsets. Different colors show results for different context lengths.

4.4 Computational cost

Table 4.2 shows the computing time (latency) in seconds for each forecasting model across different context lengths. Table 4.3 presents the peak RAM usage in GiB for each model.

Missing values indicate the model does not support that context length.

Context Length	Chronos	Sundial	TiRex	TimesFM	Moirai	Moirai (all)
24	1.5	173	104	48.2	6.5	—
96	2.4	162	107	63.9	5.2	—
168	4.0	170	91.8	65.6	5.6	—
336	6.7	169	92.9	68.8	5.8	—
512	9.5	172	94.0	70.8	7.2	110
720	—	189	91.5	73.9	12.3	—
2048	—	192	94.3	134	17.3	—
4096	—	—	103	197	20.6	—
8760	—	—	93.5	437	70.8	—
12840	—	—	98.5	635	94.6	—

Table 4.2. Computing time (latency) in seconds for each forecasting model across different context lengths. Latency represents the total time required to generate predictions for the entire test set spanning one year. Missing values indicate the model does not support that context length. Note that multivariate predictions were made on more powerful hardware.

Context Length	Chronos	Sundial	TiRex	TimesFM	Moirai	Moirai (all)
24	1.73	2.89	1.58	2.51	0.78	—
96	1.87	3.18	1.59	2.51	0.78	—
168	2.05	3.45	1.59	2.51	0.78	—
336	2.51	3.99	1.59	2.51	0.78	—
512	2.59	3.65	1.59	2.51	0.78	14.0
720	—	3.99	1.59	2.51	0.78	—
2048	—	6.79	1.59	2.51	0.78	—
4096	—	—	1.60	2.51	0.78	—
8760	—	—	1.62	2.51	0.78	—
12840	—	—	1.64	2.51	0.78	—

Table 4.3. Peak RAM usage in GiB for each forecasting model across different context lengths. Peak RAM represents the maximum memory consumed throughout the entire forecasting pipeline, including data preprocessing, model loading, and prediction generation. Missing values indicate that model does not support this length. Note that multivariate predictions from Moirai were done on different hardware.

Table A.1 shows the computing time (latency) in seconds for different feature subsets (covariates) using the Moirai model. Table A.2 presents the peak RAM usage in MiB for the same feature subsets. The “All” column represents using all covariates combined, while other columns show individual covariate features.

5. DISCUSSION

In this thesis, we establish x main findings regarding the utilization of FM models in predicting ED occupancy. Each of these along with their implications are discussed below.

Foundational models

X model outperforms other FM models

Y model is the most efficient - how accuracy changes with input length

- how input covariates changes accuracy
- did fine tuning increase accuracy
- are foundational models potential option for areas like this one, comparison to trained models, compare with other studies: Show chart with all the old results and new results. Compare them and keep telling the story of trained vs foundational
- limitations of the study (only one validation dataset)
- other observations and future work
- maybe connect with results or conclusion chapter?

5.1 Limitations

6. CONCLUSION

- summary of answers to the research questions, conclusion

7. REFERENCES

- [1] Sartini, M., Carbone, A., Demartini, A., Giribone, L., Oliva, M., Spagnolo, A. M., Cremonesi, P., Canale, F. and Cristina, M. L. Overcrowding in Emergency Department: Causes, Consequences, and Solutions—A Narrative Review. *Healthcare* 10.9 (Aug. 25, 2022), p. 1625. ISSN: 2227-9032. DOI: 10.3390/healthcare10091625. (Visited on 09/25/2025).
- [2] Berg, L. M., Ehrenberg, A., Florin, J., Östergren, J., Discacciati, A. and Göransson, K. E. Associations Between Crowding and Ten-Day Mortality Among Patients Allocated Lower Triage Acuity Levels Without Need of Acute Hospital Care on Departure From the Emergency Department. *Annals of Emergency Medicine* 74.3 (Sept. 1, 2019), pp. 345–356. ISSN: 0196-0644. DOI: 10.1016/j.annemergmed.2019.04.012. URL: <https://www.sciencedirect.com/science/article/pii/S0196064419303312> (visited on 09/25/2025).
- [3] Jo, S., Jin, Y. H., Lee, J. B., Jeong, T., Yoon, J. and Park, B. Emergency Department Occupancy Ratio is Associated With Increased Early Mortality. *Journal of Emergency Medicine* 46.2 (Feb. 1, 2014), pp. 241–249. ISSN: 0736-4679. DOI: 10.1016/j.jemermed.2013.05.026.
- [4] McCarthy, M. L., Zeger, S. L., Ding, R., Levin, S. R., Desmond, J. S., Lee, J. and Aronsky, D. Crowding Delays Treatment and Lengthens Emergency Department Length of Stay, Even Among High-Acuity Patients. *Annals of Emergency Medicine* 54.4 (Oct. 1, 2009), 492–503.e4. ISSN: 0196-0644. DOI: 10.1016/j.annemergmed.2009.03.006.
- [5] Tuominen, J., Pulkkinen, E., Peltonen, J., Kanninen, J., Oksala, N., Palomäki, A. and Roine, A. Forecasting emergency department occupancy with advanced machine learning models and multivariable input. *International Journal of Forecasting* 40.4 (Oct. 1, 2024), pp. 1410–1420. ISSN: 0169-2070. DOI: 10.1016/j.ijforecast.2023.12.002. URL: <https://www.sciencedirect.com/science/article/pii/S0169207023001346> (visited on 09/11/2025).
- [6] Hyndman, R. J. and Athanasopoulos, G. *Forecasting: Principles and Practice*. 2nd ed. Accessed on <current date>. Melbourne, Australia: OTexts, 2018. URL: <https://otexts.com/fpp2/>.

- [7] Asplin, B. R., Magid, D. J., Rhodes, K. V., Solberg, L. I., Lurie, N. and Camargo Jr, C. A. A conceptual model of emergency department crowding. *Annals of emergency medicine* 42.2 (2003), pp. 173–180.
- [8] Richardson, D. B. Increase in patient mortality at 10 days associated with emergency department overcrowding. *The Medical Journal of Australia* 184.5 (Mar. 6, 2006), pp. 213–216. ISSN: 0025-729X. DOI: 10.5694/j.1326-5377.2006.tb00204.x. URL: <https://www.mja.com.au/journal/2006/184/5/increase-patient-mortality-10-days-associated-emergency-department-overcrowding>.
- [9] : *Canadian Journal of Emergency Medicine* 3.2 (2001), pp. 82–84. DOI: 10.1017/S1481803500005285.
- [10] Bernstein, S. L., Aronsky, D., Duseja, R., Epstein, S., Handel, D., Hwang, U., McCarthy, M., McConnell, K. J., Pines, J. M., Rathlev, N., Schafermeyer, R., Zwemer, F., Schull, M. and Asplin, B. R. Emergency Department Crowding, Part 1: Definition and Magnitude. *Academic Emergency Medicine* 16.11 (2009), pp. 106–113. DOI: 10.1111/j.1553-2712.2009.00590.x.
- [11] Kulstad, E. B., Sikka, R., Sweis, R. T., Kelley, K. M. and Rzechula, K. H. ED overcrowding is associated with an increased frequency of medication errors. *The American Journal of Emergency Medicine* 28.3 (2010), pp. 304–309. ISSN: 0735-6757. DOI: <https://doi.org/10.1016/j.ajem.2008.12.014>. URL: <https://www.sciencedirect.com/science/article/pii/S0735675708008310>.
- [12] Bond, K., Ospina, M., Blitz, S., Afilalo, M., Sam, G., Bullard, M., Innes, G., Holroyd, B., Curry, G., Schull, M. and Rowe, B. Frequency, Determinants and Impact of Overcrowding in Emergency Departments in Canada: A National Survey. *Healthcare quarterly (Toronto, Ont.)* 10 (Feb. 2007), pp. 32–40. DOI: 10.12927/hcq.2007.19312.
- [13] Gul, M. and Celik, E. An exhaustive review and analysis on applications of statistical forecasting in hospital emergency departments. *Health Systems* 9.4 (2018). No potential conflict of interest was reported by the authors., pp. 263–284. ISSN: 2047-6965. DOI: 10.1080/20476965.2018.1547348. URL: <https://www.tandfonline.com/doi/full/10.1080/20476965.2018.1547348>.
- [14] Peter J. Brockwell, R. A. D. *Introduction to Time Series and Forecasting*. 2016. DOI: 10.1007/978-3-319-29854-2.
- [15] Bommasani, R. et al. *On the Opportunities and Risks of Foundation Models*. 2022. arXiv: 2108.07258 [cs.LG]. URL: <https://arxiv.org/abs/2108.07258>.
- [16] Raffel, C., Shazeer, N., Roberts, A., Lee, K., Narang, S., Matena, M., Zhou, Y., Li, W. and Liu, P. J. Exploring the Limits of Transfer Learning with a Unified Text-to-Text Transformer. *The Journal of Machine Learning Research* 21.140 (2020), pp. 1–67. URL: <http://jmlr.org/papers/v21/20-074.html>.
- [17] Vaswani, A., Shazeer, N., Parmar, N., Uszkoreit, J., Jones, L., Gomez, A. N., Kaiser, L. and Polosukhin, I. Attention Is All You Need. *Advances in Neural Information*

- Processing Systems*. 2017, pp. 5998–6008. URL: <https://arxiv.org/abs/1706.03762>.
- [18] LeCun, Y., Bengio, Y. and Hinton, G. Deep Learning. *Nature* 521.7553 (2015). Published online: 27 May 2015; Issue date: 28 May 2015, pp. 436–444. DOI: 10.1038/nature14539. URL: <https://www.nature.com/articles/nature14539>.
 - [19] Lim, B., Arik, S. Ö., Loeff, N. and Pfister, T. Temporal Fusion Transformer for Interpretable Multi-horizon Time Series Forecasting. *arXiv preprint arXiv:1912.09363* (2021).
 - [20] Domingos, P. A Few Useful Things to Know About Machine Learning. *Communications of the ACM* 55.10 (2012), pp. 78–87. DOI: 10.1145/2347736.2347755.
 - [21] Aksu, T., Woo, G., Liu, J., Liu, X., Liu, C., Savarese, S., Xiong, C. and Sahoo, D. GIFT-Eval: A Benchmark For General Time Series Forecasting Model Evaluation. *arXiv preprint arXiv:2410.10393* (2024).
 - [22] Ansari, A. F., Stella, L., Turkmen, C., Zhang, X., Mercado, P., Shen, H., Shchur, O., Rangapuram, S. S., Arango, S. P., Kapoor, S., Zschiegner, J., Maddix, D. C., Wang, H., Mahoney, M. W., Torkkola, K., Wilson, A. G., Bohlke-Schneider, M. and Wang, Y. *Chronos: Learning the Language of Time Series*. Nov. 4, 2024. DOI: 10.48550/arXiv.2403.07815. arXiv: 2403.07815. URL: <http://arxiv.org/abs/2403.07815> (visited on 09/11/2025).
 - [23] Salesforce AI Research. *GIFT-Eval Leaderboard: General Time Series Forecasting Model Evaluation*. <https://huggingface.co/spaces/Salesforce/GIFT-Eval>. Accessed: October 7, 2025. 2024.
 - [24] Das, A., Kong, W., Sen, R. and Zhou, Y. *A decoder-only foundation model for time-series forecasting*. Apr. 17, 2024. DOI: 10.48550/arXiv.2310.10688. arXiv: 2310.10688. (Visited on 09/11/2025).
 - [25] Woo, G., Liu, C., Kumar, A., Xiong, C., Savarese, S. and Sahoo, D. *Unified Training of Universal Time Series Forecasting Transformers*. May 22, 2024. DOI: 10.48550/arXiv.2402.02592. arXiv: 2402.02592. URL: <http://arxiv.org/abs/2402.02592> (visited on 09/11/2025).
 - [26] Salesforce AI Research. *Introducing Moirai 2.0*. <https://www.salesforce.com/blog/moirai-2-0/>. Accessed: October 7, 2025. 2025.
 - [27] Liu, Y., Qin, G., Shi, Z., Chen, Z., Yang, C., Huang, X., Wang, J. and Long, M. *Sundial: A Family of Highly Capable Time Series Foundation Models*. May 30, 2025. DOI: 10.48550/arXiv.2502.00816. arXiv: 2502.00816. URL: <http://arxiv.org/abs/2502.00816> (visited on 09/11/2025).
 - [28] Auer, A., Podest, P., Klotz, D., Böck, S., Klambauer, G. and Hochreiter, S. *TiRex: Zero-Shot Forecasting Across Long and Short Horizons with Enhanced In-Context Learning*. May 29, 2025. DOI: 10.48550/arXiv.2505.23719. arXiv: 2505.23719. URL: <http://arxiv.org/abs/2505.23719> (visited on 09/11/2025).

- [29] Beck, M., Pöppel, K., Spanring, M., Auer, A., Prudnikova, O., Kopp, M. K., Klambauer, G., Brandstetter, J. and Hochreiter, S. xLSTM: Extended Long Short-Term Memory. *The Thirty-eighth Annual Conference on Neural Information Processing Systems*. Nov. 2024.
- [30] Ke, G., Meng, Q., Finley, T., Wang, T., Chen, W., Ma, W., Ye, Q. and Liu, T.-Y. LightGBM: A Highly Efficient Gradient Boosting Decision Tree. *Advances in Neural Information Processing Systems*. Vol. 30. 2017, pp. 3146–3154. URL: <https://proceedings.neurips.cc/paper/2017/file/6449f44a102fde848669bdd9eb6b76fa-Paper.pdf>.

APPENDIX A: COMPUTATIONAL COSTS FOR
MULTIVARIATE FORECASTS

Feature Subset	Context Length		
	256	512	1024
All	43.3	110	370
Beds	5.3	5.6	9.9
Calendar	5.3	5.4	5.6
Events	5.2	5.2	5.4
Holiday	5.3	5.4	7.3
Hour	5.5	5.8	9.9
Month	5.3	5.4	5.5
TA	5.5	7.8	15.6
Traffic	7.4	14.1	39.2
Trends	5.1	5.2	5.3
Weather	7.3	7.5	8.2
Website	5.2	5.2	5.4
Weekday	5.3	5.6	6.6

Table A.1. Computing time (latency) in seconds for different feature subsets across context lengths (Moirai model). "All" represents all covariates combined, while other columns represent individual covariate features.

Feature Subset	Context Length		
	256	512	1024
All	1030	3730	14300
Beds	68.4	108	252
Calendar	56.5	66.1	90.0
Events	53.5	53.8	54.9
Holiday	63.9	86.5	164
Hour	70.8	112	263
Month	59.6	71.3	114
TA	84.4	171	498
Traffic	146	400	1400
Trends	53.5	54.3	55.3
Weather	62.4	78.9	144
Website	53.9	55.8	64.9
Weekday	54.8	61.0	78.9

Table A.2. Peak RAM usage in MiB for different feature subsets across context lengths for Moirai model. "All" represents all covariates combined, while other columns represent covariate subsets.